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Design and evaluation of a product analytics framework for a wearable health tracking application

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Abstract

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This thesis presents the design and evaluation of a product analytics framework for a wearable health tracking application. The goal of the study is to support data-driven decision-making through structured data collection and analysis.

The framework includes key product metrics (activation, retention, engagement, conversion), an event tracking model, and a data architecture. To demonstrate its applicability, synthetic data was used and analyzed with SQL and analytical tools.

The results show that the proposed approach helps to better understand user behavior, establish a structured approach to working with data, and can be used to improve the product. The framework can be applied to various digital products.

Keywords: product analytics, wearable applications, user behavior, data analytics, analytics framework

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Tässä opinnäytetyössä esitellään tuoteanalytiikan viitekehysten suunnittelu ja arviointi puettaviin laitteisiin perustuvaa terveyden seurantaan tarkoitettua sovellusta varten. Työn tavoitteena on tukea datalähtöistä päätöksentekoa rakenteellisen datan keruun ja analyysin avulla.

Viitekehys sisältää keskeiset tuoteanalytiikan mittarit (activation, retention, engagement, conversion), tapahtumaseurannan mallin sekä data-arkkitehtuurin. Sovellettavuuden osoittamiseksi käytettiin synteettistä dataa, jota analysoitiin SQL:n ja analytiikkatyökalujen avulla.

Tulokset osoittavat, että ehdotettu lähestymistapa auttaa ymmärtämään paremmin käyttäjien käyttäytymistä, luomaan rakenteisen lähestymistavan datan käsittelyyn ja sitä voidaan hyödyntää tuotteen kehittämisessä. Viitekehys on sovellettavissa erilaisiin digitaalisiin tuotteisiin.

Avainsanat: tuoteanalytiikka, puettavat sovellukset, käyttäjäkäyttäytyminen, data-analytiikka, analytiikan viitekehys

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List of Abbreviations

ELT: Extract, Load, Transform. A data processing approach where raw data is first loaded into a data warehouse and then transformed.

BI: Business Intelligence. Technologies and practices used for analyzing data and presenting actionable information to support decision-making.

dbt: Data Build Tool. A transformation framework that enables analytics engineers to define, test, and document data models in SQL.

DAG: Directed Acyclic Graph. A graphical representation of dependencies between data transformation models.

WAU: Weekly Active Users. The number of unique users who interact with a product within a seven-day period.

WIAU: Weekly Insight Active Users. The number of unique users who interact with the core feature (insight view) within a seven-day period.

KPI: Key Performance Indicator. A measurable value used to evaluate the success of a product or business objective.

NSM: North Star Metric. The primary metric that best captures the core value delivered by a product.

LTV: Lifetime Value. The estimated total revenue generated by a user over their entire relationship with a product.

API: Application Programming Interface. A set of rules and protocols that allows different software systems to communicate with each other.

1 Introduction

In recent years, wearable devices and mobile health applications have become part of everyday life (Piwek et al. 2016; Juniper Research 2013). Smartwatches, rings, and fitness trackers collect data on sleep, physical activity, and key physiological indicators. Users increasingly rely on these digital tools to monitor their health and track lifestyle changes.

However, the real value of such solutions lies not only in measurement accuracy, but also in the application itself. It is through the app that users access their data, view interpretations, and receive recommendations. The interface design, metric structure, and presentation of information strongly influence how often the product is used.

These applications generate large volumes of data, including both physiological time-series and detailed records of user behavior within the app (Kleppmann 2017). Yet product teams often receive only basic summary statistics and lack structured tools to understand which features deliver real value and which remain unused.

As data volumes continue to grow, a systematic approach to product analytics becomes essential. Without clear event definitions, well-defined metrics, and consistent interpretation rules, it is difficult to measure engagement, identify problems, and make informed product decisions (Provost & Fawcett 2013).

This thesis focuses on the design and evaluation of a product analytics framework for digital health applications connected to wearable devices. The study aims to develop a practical analytical structure that supports in-depth user behavior analysis and enables data-driven decision-making in product development.

2 Theoretical Framework

2.1 Product Analytics Theory

Product analytics is a branch of data analysis focused on understanding how users behave inside a digital product and using those insights to guide its development (Croll & Yoskovitz 2013). At its core, it examines the ways people interact with different features and functionalities.

Unlike traditional business analytics, which primarily tracks financial and operational performance, product analytics centers on user experience and the logic of product usage. Every user action is viewed as a measurable event that can be captured, stored, and later analyzed.

The foundation of this approach is an event-driven data model. User interactions are recorded as individual events containing the action type, timestamp, and relevant context. This structure allows teams to reconstruct user journeys and observe changes in behavior over time (Fisher et al. 2012).

The primary goal of product analytics is to uncover behavioral patterns that drive product growth. This involves understanding how users discover and adopt the product, which features deliver real value, and which factors influence continued use or churn.

Product analytics is tightly linked to the data-driven mindset: product decisions rely on quantitative evidence, and new ideas are validated through measurable metrics rather than assumptions (Provost & Fawcett 2013; Kohavi, Tang & Xu 2020).

In health applications connected to wearable devices, the data environment becomes significantly more complex. In addition to standard in-app events, teams must also work with physiological sensor data. The result is a multi-layered dataset that combines behavioral logs with time-series physiological information.

Overall, product analytics rests on three core principles: event-based data modeling, a strong focus on user value, and the use of analytics to inform product decisions. These principles serve as the foundation for designing meaningful metrics and for analyzing user retention and engagement.

2.2 Metrics Frameworks

Effective product analytics depends not only on collecting metrics, but on organizing them into a clear and coherent structure. When metrics remain isolated, it is difficult to understand how user behavior connects with overall product development.

One widely used model is the AARRR framework (Acquisition, Activation, Retention, Revenue, Referral), which structures the user lifecycle into measurable stages (McClure 2007; Croll & Yoskovitz 2013).

Another popular approach is the North Star Metric (NSM), which identifies a single core metric reflecting the primary value delivered to users (Croll & Yoskovitz 2013). Supporting indicators help explain variations in that central metric.

The idea of linking metrics through cause-and-effect relationships is conceptually aligned with broader strategic performance frameworks such as the Balanced Scorecard (Kaplan & Norton 1996).

The most suitable metric framework always depends on the product type and its business model. In applications linked to wearable devices, it is essential to track both in-app user behavior and the depth of engagement with personal health data and insights.

In this thesis, a hierarchical metric structure is adopted. It centers on one key product metric supported by a set of related indicators that reveal the underlying drivers of user behavior.

2.3 Retention and Engagement

Retention and engagement serve as core indicators of a product's long-term viability. They reveal whether users form a lasting habit of regular use (Gupta & Kim 2007).

Retention shows the percentage of users who continue interacting with the product after a given period. It is typically measured through cohort analysis, which groups users by their first interaction date and tracks their activity over time (Fader & Hardie 2010).

Engagement, in turn, reflects the depth and frequency of user interaction. Common engagement metrics include session frequency, use of key features, and responses to notifications or prompts.

Engagement is closely tied to perceived value: the more useful users find the product's features, the more often they return to them (Perski et al. 2017).

The relationship between retention and engagement works in both directions. Strong initial engagement raises the chances of users coming back, while sustained retention gives them time to develop deeper usage habits.

In wearable-connected health applications, simply receiving sensor data does not ensure active engagement. Therefore, analytics must go beyond counting app openings and instead measure meaningful interactions with reports, recommendations, and data visualizations.

For these reasons, retention and engagement lie at the heart of product analytics and serve as the foundation for building any effective analytical framework.

3 Design of the Product Analytics Framework

This section presents the design of a product analytics structure for a digital application integrated with a wearable health device. The goal is to build an event tracking system and define a metric structure for analyzing user behavior, engagement, retention, and monetization.

The framework is based on an event-driven data model that records meaningful user actions and enables their interpretation within product analytics (Fisher et al. 2012).

From a data architecture perspective, event-based modeling aligns with dimensional modeling principles commonly used in analytical data warehouses (Kimball & Ross 2013).

3.1 Event Tracking Model

The event tracking model is the core of the developed framework. Each meaningful user action is recorded as a structured event.

Each event contains:

- user_id
- timestamp
- event_name
- event_properties
- technical attributes (app version, device type)

This model allows:

- analysis of user journeys
- funnel construction
- calculation of retention and engagement
- measurement of conversion
- evaluation of product changes

For a wearable application, two interaction types are considered:

1. Behavioral actions inside the interface
2. Interaction with health analytics and personal insights

This section presents the design of a product analytics structure for a digital application integrated with a wearable health device. The goal is to build an event tracking system and define a metric structure for analyzing user behavior, engagement, retention, and monetization.

The framework is based on an event-driven data model that records meaningful user actions and enables their interpretation within product analytics.

3.1.1 Definition of User Events

Key events were defined to reflect interaction with core features.

1. Launch and activation

- app_install
- app_open
- onboarding_complete

2. Health data views

- sleep_view
- readiness_view
- activity_view
- trend_view

3. Analytical interaction

- insight_view
- recommendation_click
- notification_click

4. Subscription events

- paywall_view
- subscription_start
- subscription_renew
- subscription_cancel
- subscription_expired

These events cover activation, engagement, retention, and monetization.

3.1.2 Retention

Retention is commonly evaluated using cohort-based approaches that track user activity over time (Fader & Hardie 2010).

Retention is defined as the percentage of users who perform a meaningful action n days after activation.

Day 0 is defined as onboarding_complete.

C_0 — users who completed onboarding on day t_0

A_n — users from this cohort active on day $t_0 + n$

$$\text{Retention}_n = |A_n| / |C_0| \quad (1)$$

A user is considered active if at least one of the following events is recorded:

- app_open
- sleep_view
- readiness_view
- insight_view

The framework uses fixed-day retention and cohort analysis.

In wearable applications, sensor data is generated automatically. Therefore, retention is measured through active interaction with analytics rather than passive data generation.

3.1.3 Engagement

Retention is commonly evaluated using cohort-based approaches that track user activity over time (Fader & Hardie 2010).

Engagement reflects the frequency and depth of product usage.

The following events are included:

- app_open
- sleep_view
- readiness_view
- activity_view
- insight_view
- recommendation_click
- notification_click

Engagement Frequency

Engagement_frequency = Number of active days / Total days in the period (2)

Engagement Depth

Engagement_depth = Number of deep interaction events/Number of sessions (3)

A session is defined as a sequence of events separated by no more than 30 minutes of inactivity.

Engagement Score

$$\text{Engagement Score} = w_1 \cdot E_1 + w_2 \cdot E_2 + w_3 \cdot E_3 \quad (4)$$

Higher weights are assigned to deep interaction events.

In wearable applications, engagement is defined by active analysis and interpretation of health data rather than passive data collection.

3.1.4 Monetization and Subscription

Engagement reflects the frequency and depth of product usage and has been conceptualized as a multidimensional construct (O'Brien & Toms 2008; Perski et al. 2017).

The product uses a subscription-based monetization model.

The following events are tracked:

- paywall_view
- subscription_start
- subscription_renew
- subscription_cancel
- subscription_expired

Conversion to Paid

$$\text{Conversion_to_paid} = \text{subscription_start} / \text{paywall_view} \quad (5)$$

$$\text{Paid_conversion_total} = \text{Subscribers} / \text{Users who completed onboarding} \quad (6)$$

Subscription Churn

$$\text{Subscription_churn_rate} = \text{subscription_cancel} / \text{Active subscribers} \quad (7)$$

Revenue Metrics

$$\text{ARPU} = \text{Total revenue} / \text{Active users} \quad (8)$$

$$\text{ARPPU} = \text{Total revenue} / \text{Paying users} \quad (9)$$

The framework assumes analysis of the relationship between engagement and subscription probability.

3.1.5 North Star Metric

Subscription-based monetization models are widely used in digital products and require careful measurement of conversion and churn dynamics (Kohavi, Tang & Xu 2020).

The North Star Metric represents the core value of the product.

It must:

- reflect user value
- correlate with long-term retention
- be measurable from event data
- support product growth

In this thesis, the selected NSM is:

Weekly Insight Active Users (WIAU)

WIAU =

Number of unique users who performed insight_view or recommendation_click at least once within 7 days (10)

This metric reflects regular interaction with analytics rather than simple app openings.

Supporting metrics include:

- activation rate
- D7 and D30 retention
- engagement depth
- subscription conversion

3.2 Core Product Metrics

This section defines the core product metrics and the data structure required for their calculation. The metrics are based on the previously defined event model and the selected North Star Metric.

The goal is to ensure consistency between events, metrics, and the data architecture.

3.2.1 Activation

Activation represents the first meaningful action that confirms the user has received initial product value.

In this application, activation is defined as:

- completion of onboarding
- first intentional view of personal health data

Activation events:

- onboarding_complete
- first_sleep_view
- first_readiness_view

A user is considered activated if, within 24 hours after installation:

- onboarding_complete is completed
- at least one analytics view is recorded

Activation Rate =

Number of activated users / Number of users who installed the app

This metric evaluates the effectiveness of the initial user experience.

3.2.2 Event Schema

The event schema defines how user actions are stored.

Each event includes:

- event_id
- user_id

- event_name
- event_timestamp
- event_properties
- device_type
- app_version

Additional attributes may include:

- acquisition_source
- country
- subscription_type

This structure ensures flexibility and scalability of the analytical system.

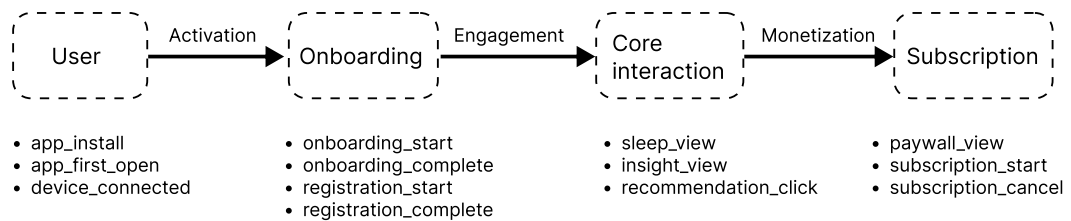


Figure 1. User journey logic and corresponding event groups.

Figure 1 presents a simplified user journey and corresponding event groups.

3.2.3 Data Structure Design

A relational data model is proposed for storing and analyzing events.

Main tables:

1. users
2. events
3. subscriptions
4. sessions

Users table:

- user_id (PK)
- registration_date
- country

- acquisition_source

Events table:

- event_id (PK)
- user_id (FK)
- event_name
- event_timestamp
- event_properties

Subscriptions table:

- subscription_id
- user_id
- start_date
- end_date
- status

Sessions table:

- session_id
- user_id
- session_start
- session_end

This structure enables calculation of retention, engagement, North Star, and monetization metrics.

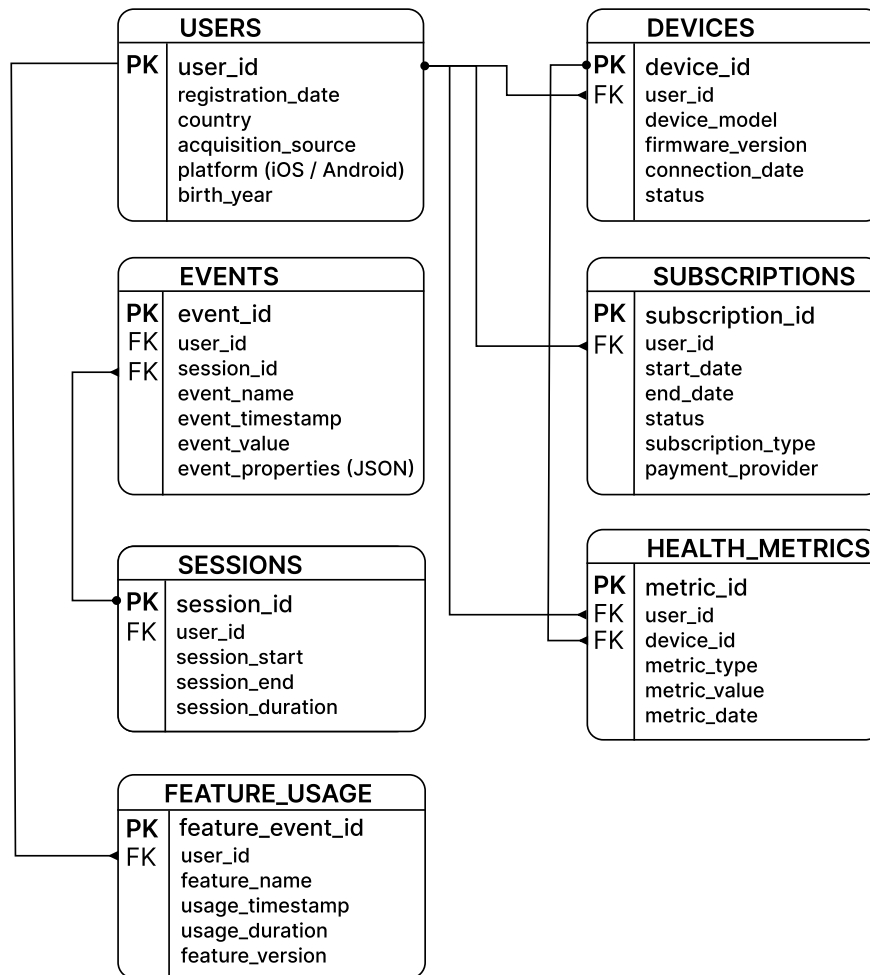


Figure 2. ER diagram of the developed analytical data model.

Figure 2 shows the ER diagram of the designed data model. The central entity is Users, which has one-to-many relationships with Events, Sessions, and Subscriptions. The Events table is the primary source for calculating product metrics.

3.3 Funnel and User Journey Modeling

Funnel analysis is used to examine sequences of user actions and identify stages where conversion decreases. Funnels are built based on fixed event sequences that represent key stages of the user lifecycle.

This study includes three main funnels:

1. Activation

2. Engagement
3. Monetization

3.3.1 Activation Funnel

The goal of the activation funnel is to measure the percentage of users who reach their first meaningful interaction with analytics.

Stages:

1. app_install
2. app_first_open
3. onboarding_complete
4. device_connected
5. first_sleep_view or insight_view

Activation Funnel Conversion =

Users who reached the final stage / Users who installed the app

This funnel helps identify weaknesses in the initial user experience.

Stage-to-stage conversion is calculated as:

Users who reached the current stage / Users who reached the previous stage

The final stage is insight_view, as it reflects the first analytical value received by the user.

Figure 3 shows the sequence of events forming the activation process.



Figure 3. User activation funnel.

3.3.2 Engagement Funnel

The goal of the engagement funnel is to determine whether users move from basic usage to regular interaction.

Stages:

1. app_first_open
2. sleep_view
3. insight_view
4. recommendation_click
5. repeated interaction within 7 days

This funnel reflects usage depth and is directly connected to the North Star Metric (WIAU).

3.3.3 Monetization Funnel

The monetization funnel measures the transition from product usage to paid subscription.

Stages:

1. insight_view
2. paywall_view
3. subscription_start
4. subscription_renew

$$\text{Conversion_to_paid} = \text{subscription_start} / \text{paywall_view} \quad (11)$$

This funnel allows analysis of how engagement influences subscription probability.

The final stage is subscription_renew, since renewal reflects confirmed long-term product value.

Paid retention is calculated as:

$$\text{subscription_renew} / \text{subscription_start}$$

Figure 4 shows the monetization funnel.

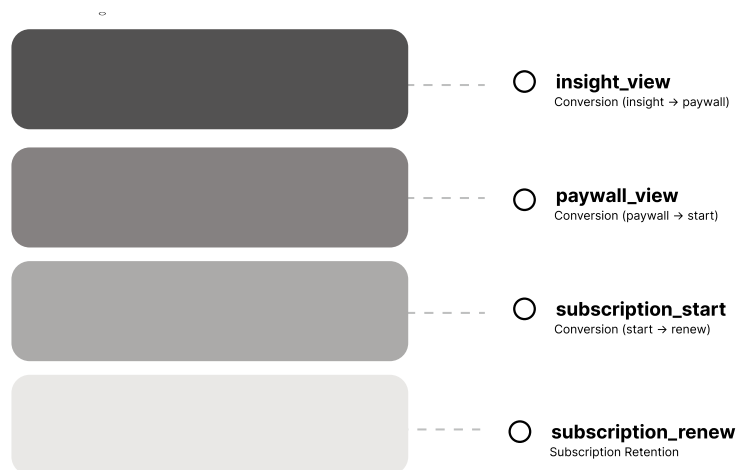


Figure 4. Monetization funnel.

User Path Analysis

In addition to fixed funnels, the model includes event sequence analysis. This method identifies the most common interaction paths.

Examples:

- sleep_view → insight_view → recommendation_click
- notification_click → insight_view
- insight_view → paywall_view

Path analysis helps detect behavioral patterns and improve interface structure.

3.4 Data Architecture

The data architecture is based on the event-driven model and supports collection, storage, processing, and analysis of user actions and wearable device data.

The architecture includes four layers:

1. Data Collection
2. Storage
3. Processing and Modeling
4. Analytics and Visualization

3.4.1 Data Collection

Data is collected at the level of the mobile application and the connected wearable device.

Data sources include:

- user interface events (app_open, insight_view, etc.);
- device metrics (sleep, activity, readiness);
- subscription events (subscription_start, subscription_cancel).

Each event is generated in a structured JSON format and sent to the analytics server.

3.4.2 Data Storage

Events are stored in a centralized data warehouse.

The main tables include:

- events
- users
- sessions
- subscriptions

- health_metrics

The storage layer maintains historical data and enables cohort analysis.

3.4.3 Data Processing and Modeling

During the processing stage, data undergoes:

- duplicate removal;
- format validation;
- time-based aggregation;
- calculation of derived metrics.

Based on the event table, analytical models are created to calculate:

- retention;
- engagement;
- conversion;
- the North Star Metric;
- subscription retention.

3.4.4 Analytics Layer

The final layer provides visualization and interpretation of metrics.

Metrics can be presented as:

- dashboards;
- cohort tables;
- funnels;
- time series charts.

The architecture follows a multi-layer analytical pipeline approach. Events generated by the application and wearable device are stored in the raw data layer, transformed and aggregated, and then used to calculate product metrics and build analytical views.

Figure 5 presents the analytical data processing architecture.



Figure 5. Analytical data processing pipeline architecture.

3.5 Data Quality and Governance

Within the developed framework, special attention is given to data quality and management of the analytical infrastructure. The reliability of product metrics depends directly on the correctness of event collection, storage, and processing.

3.5.1 Event Data Quality

Since the architecture is based on an event-driven model, the main risks include:

- duplicate events;
- lost events;
- incorrect timestamps;
- schema violations;
- incomplete user identifiers.

To reduce these risks, the framework includes:

- mandatory fields (user_id, event_name, event_timestamp);
- format validation during data collection;
- control of event_id uniqueness;
- validation of session_id consistency.

3.5.2 Event Schema Governance

Analytical stability requires a fixed schema and consistent naming rules.

The framework includes:

- a centralized event registry;
- documentation of event_properties;
- restriction of arbitrary event name changes;
- compatibility control when adding new fields.

This prevents errors in retention, conversion, and North Star calculations.

3.5.3 Metric Integrity Control

After data transformation, metrics are validated through:

- comparison of aggregates with raw data;
- anomaly detection;
- monitoring of missing value rates;
- validation of cohort calculations.

Regular checks reduce the risk of incorrect management decisions.

3.5.4 Access Control and Privacy

Since the application processes health-related data, the architecture must comply with privacy requirements.

The framework includes:

- pseudonymization of user identifiers;
- role-based access control;
- separation of personal data from analytical tables.

These measures are especially important under European data protection regulation (GDPR).

3.5.5 Naming Convention

To ensure consistency, a unified naming convention is defined.

Lack of standardization can lead to:

- duplicated events;
- incorrect metric calculations;
- broken cohort analysis;
- BI reporting errors.

Event Naming Principles

- Use snake_case format.
- The event name must reflect a user action.
- Events describe interaction, not interface elements.

Examples:

1. app_install
2. onboarding_complete

3. device_connected
4. insight_view
5. subscription_start
6. subscription_renew

Parameter Naming Principles

Standardized parameters include:

- subscription_type
- plan_price
- device_model
- insight_type
- feature_version

The following are not allowed:

- spaces;
- mixed casing;
- ambiguous abbreviations.

Table Naming

Storage tables use plural form:

- users
- events
- sessions
- subscriptions
- health_metrics

Aggregated tables use a prefix:

- agg_user_retention
- agg_subscription_metrics
- agg_north_star

Schema Versioning

When the event structure changes, schema versioning is applied to maintain compatibility with previously calculated metrics.

3.5.6 Monitoring of the Analytical System

To ensure stable operation of the analytical infrastructure, the framework includes a monitoring system.

Monitoring aims to:

- detect event collection errors;
- validate transformation processes;
- identify metric anomalies;
- prevent distortion of management indicators.

Event Flow Monitoring

The system checks:

- event volume over time;
- distribution by event type;
- presence of mandatory fields;
- error rates during data ingestion.

Sudden drops or structural changes may indicate technical issues.

Transformation Validation

At the processing layer, the following are monitored:

- successful execution of transformations;
- correctness of aggregations;
- consistency between aggregated and raw data;

- error logging.

Metric Monitoring

Key metrics (retention, conversion, North Star, subscription retention) are regularly reviewed for anomalies.

Monitoring includes:

- detection of sudden spikes;
- comparison with historical data;
- validation of cohort stability.

Monitoring builds trust in data. Without control mechanisms, even a well-designed architecture may lead to incorrect product decisions.

4 Implementation

In the practical part of this thesis, a synthetic database was created to demonstrate the functionality of the developed product analytics framework. The dataset simulates user behavior in a digital health monitoring application.

The use of synthetic data is обусловлено privacy requirements and the lack of access to real user data. The generated dataset replicates a typical event-based structure of a digital product.

The following parameters were defined during data generation:

- 1000 users
- observation period — 120 days
- three behavioral types (casual, regular, power users)
- different probabilities of activation, engagement, and subscription
- monthly subscription model with renewal
- daily wearable metrics (sleep, activity, readiness)

The behavioral model was designed to reflect differences in engagement level and subscription probability. This allows demonstration of key product metrics, including retention, conversion, North Star metric, and monetization indicators.

Data generation was implemented in Python using a probabilistic behavioral model.

The synthetic dataset is further used for SQL transformations, analytical model creation, and metric calculation in Snowflake.

4.1 SQL Transformations

4.1.1 Validation of Raw Data

At the first stage, the correctness of data loaded into the raw layer was verified. The objective was to ensure structural integrity, absence of missing values and duplicates, and consistency with the defined observation period.

Since the practical part uses a synthetic dataset modeling 120 days of user activity, it was necessary to confirm that the data was correctly loaded into Snowflake.

Missing Value Check

```
SELECT COUNT(*)
FROM raw.events
WHERE user_id IS NULL
      OR event_name IS NULL
      OR event_timestamp IS NULL;
```

Listing 1. SQL query for validation of mandatory fields in the raw.events table.

The query returned zero records, confirming that no missing values exist in mandatory fields.

Duplicate Check

```
SELECT event_id, COUNT(*)
FROM raw.events
GROUP BY event_id
HAVING COUNT(*) > 1;
```

Listing 2. SQL query for detection of duplicate event_id values.

No duplicate records were found. Each event appears only once in the dataset.

Timestamp Validation

```
SELECT MIN(event_timestamp), MAX(event_timestamp)
FROM raw.events;
```

Listing 2. SQL query for detection of duplicate event_id values.

The returned date range corresponds to the defined 120-day observation period.

Data Volume Check

```
SELECT COUNT(*) FROM raw.users;  
SELECT COUNT(*) FROM raw.sessions;  
SELECT COUNT(*) FROM raw.events;  
SELECT COUNT(*) FROM raw.feature_usage;  
SELECT COUNT(*) FROM raw.subscriptions;  
SELECT COUNT(*) FROM raw.health_metrics;
```

Listing 4. SQL queries for validation of record counts in raw tables.

The number of records matches the expected volume generated during the synthetic data creation phase.

Conclusion

The validation confirms that the data is correctly loaded into Snowflake, contains no missing values or duplicates, and corresponds to the defined temporal structure. The dataset is ready for transformation.

4.1.2 Data Preparation and Analytical Layer Formation

After validating the raw data, an analytical layer was created in the analytics schema. Separating raw and analytical layers follows the principles of multi-layer data warehouse architecture and prevents direct manipulation of source data.

```
CREATE OR REPLACE SCHEMA analytics;
```

Listing 5. Creation of the analytics schema.

Session Preparation

A cleaned session model was created to calculate session duration and remove incorrect records.

```
CREATE OR REPLACE VIEW analytics.sessions_clean AS
SELECT
    session_id,
    user_id,
    session_start,
    session_end,
    DATEDIFF('minute', session_start, session_end) AS session_duration
FROM raw.sessions
WHERE session_end > session_start;
```

Listing 6. Creation of the cleaned sessions view.

This transformation removes sessions with invalid time intervals and enables engagement depth calculations.

Daily Activity Model

To support retention and active user calculations, a daily activity aggregation was created.

```
CREATE OR REPLACE VIEW analytics.daily_activity AS
SELECT
    user_id,
    DATE(session_start) AS activity_date,
    COUNT(*) AS sessions_count
FROM analytics.sessions_clean
GROUP BY 1,2;
```

Listing 7. Creation of daily user activity view.

This model represents whether a user was active on a specific day and is later used for retention and DAU/WAU calculations.

Insight Events Model (North Star Base)

To calculate the North Star metric (WIAU), a separate analytical model was created containing insight interactions.

```
CREATE OR REPLACE VIEW analytics.insight_events AS
SELECT
    user_id,
    event_timestamp,
    DATE(event_timestamp) AS event_date
FROM raw.events
WHERE event_name = 'insight_view';
```

Listing 8. Creation of insight events view for North Star calculation.

This model isolates events representing core product value.

4.1.3 Calculation of Key Product Metrics

Based on the analytical layer, key product metrics were calculated. These metrics reflect user activity, value delivery, engagement intensity, and monetization performance.

Weekly Active Users (WAU)

WAU represents the number of unique users who had at least one session during a calendar week.

```
SELECT
    DATE_TRUNC('WEEK', session_start) AS week,
    COUNT(DISTINCT user_id) AS wau
FROM analytics.sessions_clean
GROUP BY 1
ORDER BY 1;
```

Listing 9. SQL query for Weekly Active Users calculation.

Results:

- Average WAU: 768.8
- Minimum: 94
- Maximum: 969

WAU reflects overall platform activity but does not measure value interaction.

North Star Metric — Weekly Insight Active Users (WIAU)

WIAU measures users who interacted with personalized analytics during a week.

```

SELECT
    DATE_TRUNC('WEEK', event_timestamp) AS week,
    COUNT(DISTINCT user_id) AS wiau
FROM analytics.insight_events
GROUP BY 1
ORDER BY 1;

```

Listing 10. SQL query for Weekly Insight Active Users calculation.

Results:

- Average WIAU: 692.4
- Minimum: 70
- Maximum: 882

Unlike WAU, WIAU measures users who reached analytical value.

Value Engagement Ratio

To measure how many active users reach core value, the following ratio was calculated:

$$\begin{aligned}
 \text{Value Engagement Ratio} &= \frac{WIAU}{WAU} \\
 \frac{692.4}{768.8} &= 0.901
 \end{aligned}
 \tag{12}$$

Average value: 90.1%

This indicates that more than 90% of active users interact with personalized insights.

Interpretation:

- Core value is well integrated into the product flow
- Users move from simple usage to meaningful interaction
- The choice of WIAU as North Star is justified

Conversion to Paid

Conversion to Paid measures the proportion of users who subscribed.

```
SELECT
    COUNT(DISTINCT s.user_id) * 1.0
    / COUNT(DISTINCT u.user_id) AS conversion_to_paid
FROM raw.users u
LEFT JOIN raw.subscriptions s
    ON u.user_id = s.user_id;
```

Listing 11. SQL query for subscription conversion calculation.

Result: 19.6%

Approximately one in five users converts to a paid subscription.

Subscription Retention

Subscription retention measures the percentage of users who renewed their subscription.

```
WITH first_sub AS (
    SELECT
        user_id,
        MIN(subscription_start) AS first_start
    FROM raw.subscriptions
    GROUP BY user_id
),
renewals AS (
    SELECT DISTINCT user_id
    FROM raw.subscriptions
    WHERE subscription_status = 'renewed'
)
SELECT
    COUNT(DISTINCT r.user_id) * 1.0
    / COUNT(DISTINCT f.user_id) AS subscription_retention
FROM first_sub f
LEFT JOIN renewals r
    ON f.user_id = r.user_id;
```

Listing 12. SQL query for subscription retention calculation.

Result: 74.5%

More than 70% of subscribers renew after the first period.

Cohort-Based Retention

Retention was calculated using cohort analysis grouped by registration week.

Week 1 retention (average across cohorts): 95.5%

High early retention indicates effective activation logic and fast value delivery.

Table 1. Summary of Key Metrics

Metric	Value	Interpretation
AVG_WAU	768.8	Weekly active users
AVG_WIAU	692.4	Users reaching analytical value
Value Engagement Ratio	90.1%	Share of active users reaching value
Conversion to Paid	19.6%	Subscription conversion rate
Subscription Retention	74.5%	Renewal rate
Week 1 Retention	95.5%	Early retention performance

Analytical Conclusion

The results demonstrate:

- High engagement levels
- Strong integration of core value into user journeys
- Sustainable subscription model
- High early retention

The Value Engagement Ratio (90.1%) is particularly important, as it confirms that most active users reach the main product value. This supports the validity of the event model and the selection of WIAU as the North Star Metric.

4.2 dbt Project Structure

To systematize data transformations and build the analytical layer, a modular data modeling approach was applied using dbt (data build tool).

dbt enables structuring SQL transformations as managed models that form a Directed Acyclic Graph (DAG). This approach ensures transparency of transformation logic, reproducibility of calculations, and systematic data quality control.

4.2.1 Overall Project Architecture

Within this work, the project is logically divided into three layers:

- Staging layer (stg_)
- Intermediate layer (int_)
- Mart layer (mart_)

This separation follows modern analytics engineering best practices.

1. Staging Layer

Purpose:

- cleaning raw data
- standardizing data types
- normalizing column names
- applying basic filtering

Example models:

- stg_users
- stg_events
- stg_subscriptions
- stg_sessions
- stg_health_metrics

- stg_feature_usage

At this level, data remains close to the raw structure but is already standardized and cleaned.

2. Intermediate Layer

Purpose:

- aggregation
- user cohort formation
- daily activity calculation
- preparation of retention analysis datasets

Example models:

- int_user_cohorts
- int_daily_activity
- int_cohort_sizes
- int_retention_base

At this stage, analytical logic is implemented, but final KPI aggregation is not yet performed.

3. Mart Layer

Purpose:

- business metric calculation
- KPI computation
- preparation of data for dashboards

Example models:

- mart_wau
- mart_wiau

- mart_conversion
- mart_subscription_retention
- mart_retention_summary

This layer serves as the final semantic layer consumed by BI tools and analytical interfaces.

4.2.2 DAG Logic

The model dependencies form a Directed Acyclic Graph (DAG):

RAW TABLES

↓

STAGING MODELS

↓

INTERMEDIATE MODELS

↓

MART MODELS

↓

DASHBOARDS / ANALYTICS

This structure provides:

- clear visibility of dependencies
- automatic downstream updates
- integrity control
- reproducible metric calculations

4.2.3 Data Quality Management

Within the project, standard dbt testing mechanisms can be applied:

- uniqueness tests (unique)
- not-null tests (not_null)
- relationship tests (relationships)

- accepted values checks (accepted_values)

For example:

- user_id must be unique in the users table
- subscription_id must not be NULL
- user_id in subscriptions must exist in the users table

Thus, dbt is used not only as a transformation tool but also as a structured quality control mechanism for the analytical system.

4.2.4 Justification for Using dbt

The use of dbt in this work is justified by the following factors:

1. Modularity and structured transformations
2. Support for DAG-based architecture
3. Transparency of business logic
4. Built-in metric testing capabilities
5. Alignment with modern analytics engineering practices

dbt serves as the implementation layer of the proposed product analytics framework within the data architecture.

The developed dbt project structure:

- reflects the previously defined metric model
- ensures reproducibility of calculations
- enables system scalability
- separates raw data from business metric layers

4.2.5 dbt Project Directory Structure

To ensure modularity, scalability, and transparency of dependencies, a standard layered dbt project structure was used.

Below is the logical project structure:

```

product_analytics_dbt/
├── dbt_project.yml
├── packages.yml
├── models/
│   ├── staging/
│   │   ├── stg_users.sql
│   │   ├── stg_events.sql
│   │   ├── stg_sessions.sql
│   │   ├── stg_subscriptions.sql
│   │   ├── stg_health_metrics.sql
│   │   └── stg_feature_usage.sql
│   ├── intermediate/
│   │   ├── int_user_cohorts.sql
│   │   ├── int_daily_activity.sql
│   │   ├── int_cohort_sizes.sql
│   │   └── int_retention_base.sql
│   └── marts/
│       ├── mart_wau.sql
│       ├── mart_wiau.sql
│       ├── mart_conversion.sql
│       ├── mart_subscription_retention.sql
│       └── mart_retention_summary.sql
├── tests/
│   ├── users.yml
│   ├── subscriptions.yml
│   └── events.yml
├── macros/
│   └── metric_calculations.sql
└── snapshots/
    └── subscriptions_snapshot.sql
  
```

Listing 13. dbt project structure.

Structure Justification

1. Layer Separation

staging

Contains cleaning and standardization models.

Each model corresponds to a single raw table.

intermediate

Contains aggregated datasets and calculation logic forming the basis for metric computation.

marts

Final data marts used for KPI calculation and dashboarding.

2. Tests Directory

Contains declarative data quality tests:

- identifier uniqueness
- field completeness
- relationship validation

3. Macros Directory

Allows centralized definition of reusable logic, such as:

- retention calculation
- conversion rate formulas
- active user definitions

4. Snapshots Directory

Used to track subscription changes over time (status updates, cancellations, renewals).

This is critical for accurate monetization and subscription retention analysis.

Architectural Logic

The directory structure reflects a layered analytical architecture:

RAW → STAGING → INTERMEDIATE → MART → DASHBOARD

This approach ensures:

repeatability of calculations

isolation of business logic

system scalability

maintainability of the project

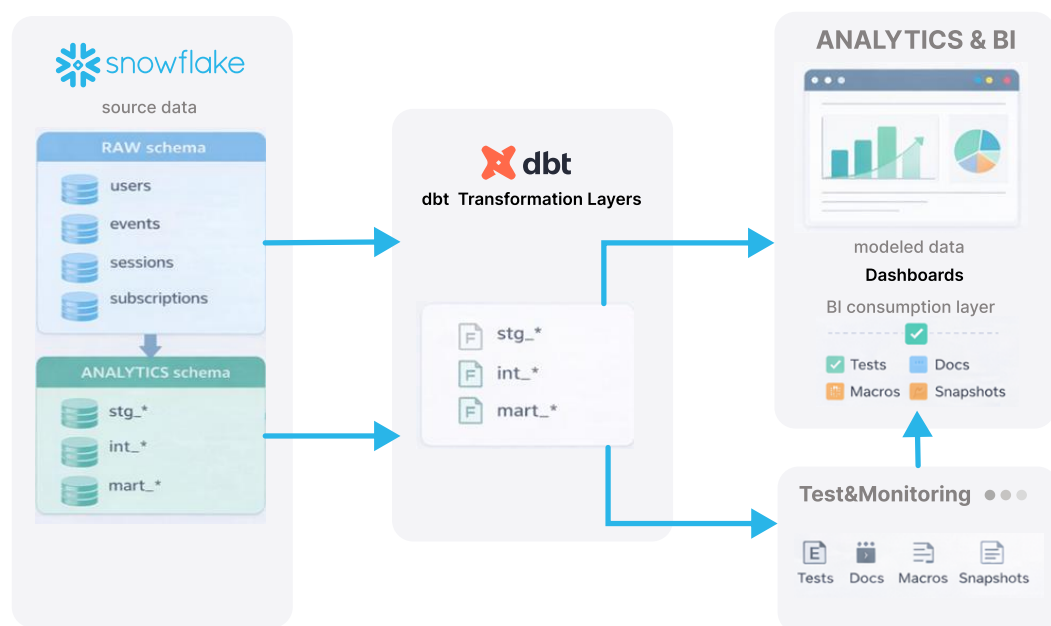


Figure 6. Architecture of the analytical pipeline based on Snowflake and dbt.

Figure 6 illustrates the logical architecture of the analytical system.

Raw event and subscription data are physically stored in the Snowflake RAW schema, representing the immutable source layer of the system.

Data transformations are implemented using dbt, which materializes structured models across three transformation layers — staging (`stg_*`), intermediate (`int_*`), and mart (`mart_*`) — within the ANALYTICS schema. This layered structure

ensures modularity, separation of concerns, and reproducibility of analytical logic.

The resulting mart models serve as a semantic data layer and are consumed by business intelligence tools for dashboarding, KPI monitoring, and decision support.

Additionally, the architecture incorporates data testing, documentation, and monitoring mechanisms provided by dbt, ensuring analytical reliability and operational robustness.

4.3 Examples of Analytical Queries and Interpretation of Results

This section presents examples of analytical queries implemented in Snowflake, along with their interpretation in the context of product analytics.

The objective is to demonstrate the practical application of the developed analytical architecture for evaluating user engagement, conversion, and behavioral factors influencing monetization.

4.3.1 Product Funnel Analysis

To evaluate the user journey, a simplified product funnel was constructed including the following key events:

- app_first_open
- insight_view
- paywall_view
- subscription_start

Based on aggregated data, the following results were obtained:

- Total number of users: 1000
- Users who started a subscription: 196
- Conversion to paid: 19.6%

This metric represents the proportion of users who transitioned from free usage to the paid subscription model.

The observed value reflects moderate monetization efficiency and confirms that the synthetic dataset was generated with realistic behavioral distribution patterns.

4.3.2 Engagement Comparison: Free vs Paid Users

To evaluate differences in engagement levels, the average number of insight_view events was calculated for two user groups.

Table 2. Results

User Type	n	Mean	SD
Free	804	40.58	18.00
Paid	196	56.47	19.86

The difference in means:

$$\Delta = 15.89 \quad (13)$$

To test statistical significance, a Welch t-test was performed.

The obtained value $t \approx 10.25$ ($p < 0.001$) indicates a statistically significant difference between the two groups.

Thus, users who subscribe demonstrate substantially higher engagement levels compared to free users.

4.3.3 Regression Analysis: Impact of Engagement on Payment Probability

To evaluate the predictive relationship between engagement and subscription probability, a logistic regression model was constructed.

- Dependent variable: subscription status (0 = Free, 1 = Paid)
- Independent variable: number of insight_view events

The estimated model:

$$\text{logit}(p) = -4.16 + 0.055 \cdot \text{insight_views} \quad (14)$$

The engagement coefficient is positive ($\beta = 0.055$), indicating a direct relationship between interaction frequency and payment probability.

The odds ratio:

$$e^{0.055} \approx 1.056 \quad (15)$$

This means that each additional insight view increases the odds of subscription by approximately 5.6%.

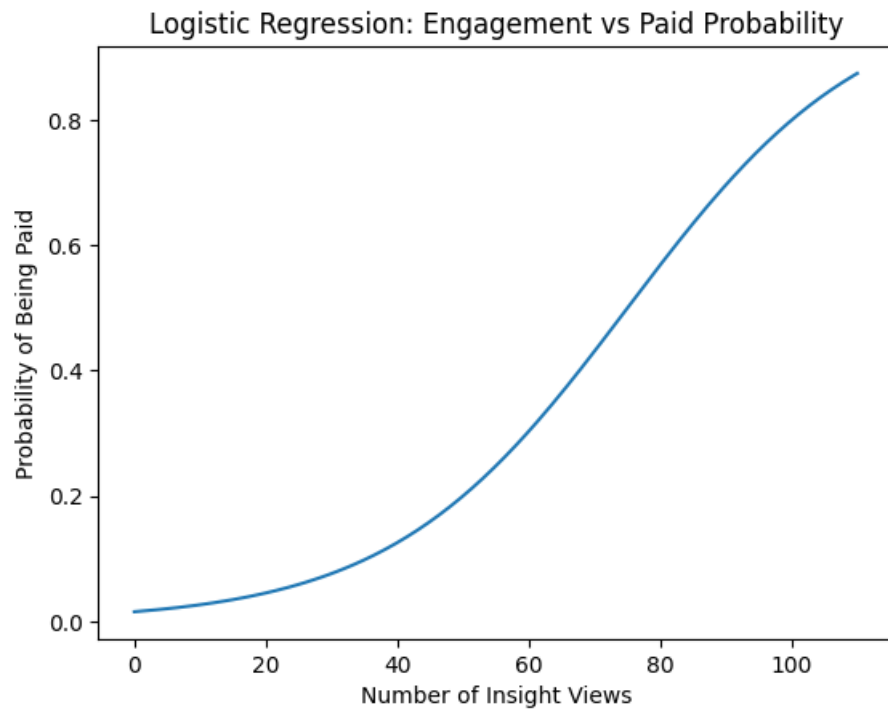


Figure 7. Logistic regression curve showing the relationship between engagement and subscription probability.

The logistic curve demonstrates an S-shaped relationship:

- At low engagement levels, subscription probability remains minimal.
- After reaching a behavioral threshold, the probability of conversion increases more rapidly.

This confirms the presence of a behavioral tipping point in monetization dynamics.

4.3.4 Interpretation of Results

The obtained results allow the following conclusions:

- User engagement is statistically significantly associated with monetization.
- The number of insight views can be considered a behavioral indicator of payment probability.
- Engagement metrics may be used as triggers for personalized subscription offers.

- The developed analytical architecture supports not only descriptive metrics but also predictive modeling.

Thus, the practical implementation demonstrates that the proposed product analytics framework supports descriptive, explanatory, and predictive analytics within a unified architectural system.

4.4 Managerial Interpretation

A comprehensive review of the dashboards allows the following managerial conclusions:

The depth of core feature usage is a key driver of monetization.

Engagement metrics can serve as early indicators of future conversion.

Monitoring engagement ratios and conversion dynamics enables evaluation of product changes.

The developed framework supports the transition from descriptive analytics to explanatory and predictive decision-making.

Thus, visualization of metrics in analytical dashboards demonstrates the practical applicability of the proposed product analytics architecture and its ability to support data-driven product management.

4.4.1 Dynamics of Active Audience and Core Feature Usage

To evaluate user activity, a dashboard was built showing the dynamics of Weekly Active Users (WAU) and Weekly Insight Active Users (WIAU) over the analyzed period.

WAU reflects the total number of unique users active during a week, while WIAU represents users who interacted with the core product function — `insight_view`.

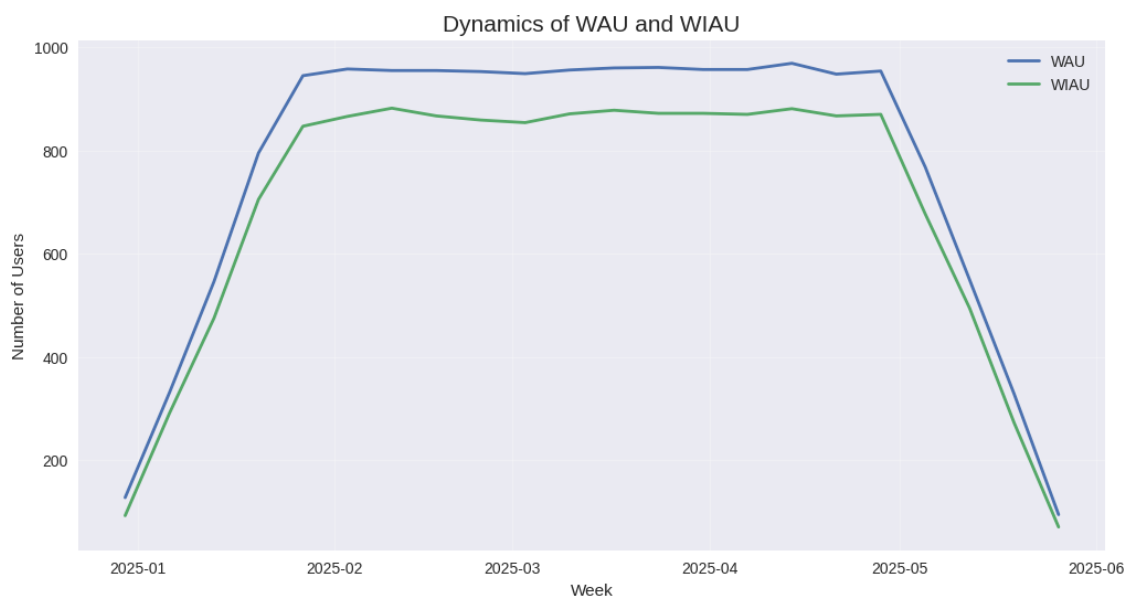


Figure 8. Dynamics of WAU and WIAU

Interpretation

As shown in Figure 8, after the initial growth phase, WAU stabilizes at approximately 950–970 users per week. This indicates the formation of a stable active audience.

WIAU demonstrates a similar trend but consistently remains below WAU, reflecting the proportion of users who actively use the core functionality.

The gap between WAU and WIAU allows evaluation of value consumption depth. During the observed period, approximately 85–90% of active users interact with the `insight_view` function, indicating high product relevance.

The decrease at the end of the period is related to an incomplete observation week and does not indicate structural decline in activity.

4.4.2 Subscription Conversion Dynamics

To analyze monetization trends more accurately, weekly subscription conversion was visualized using a 3-week moving average.

Smoothing reduces short-term fluctuations and reveals the overall trend.

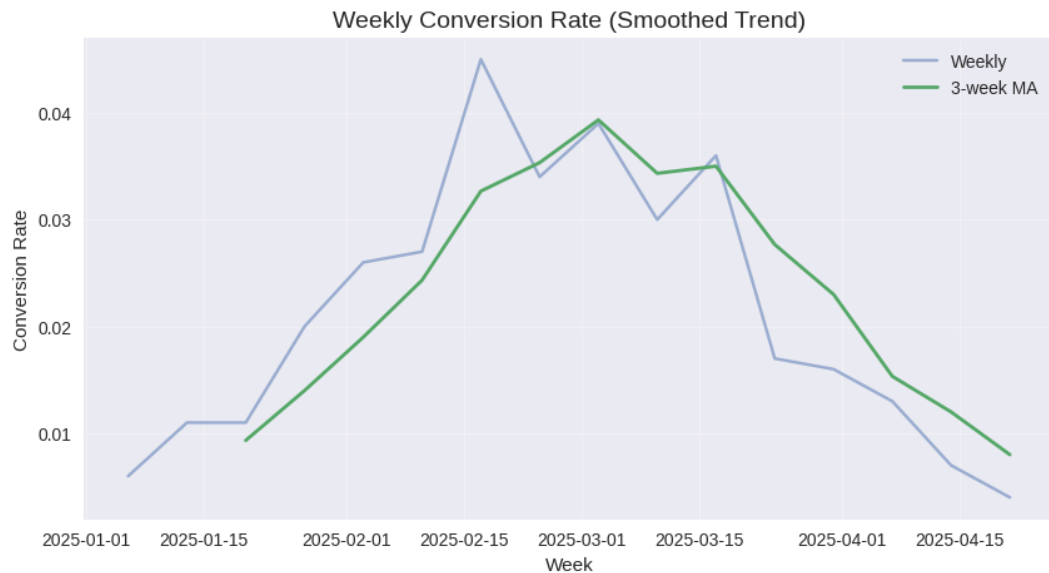


Figure 9. Weekly Subscription Conversion (Smoothed Trend)

Interpretation

As shown in Figure 9, conversion demonstrates steady growth at the beginning of the analyzed period — from below 1% to peak values of approximately 4–4.5%.

The smoothed trend confirms gradual improvement in monetization efficiency, likely influenced by accumulated user engagement.

After reaching its maximum, conversion begins to decline. The smoothed curve indicates that the decrease is consistent rather than caused by random weekly fluctuations.

This dynamic supports the conclusions of Section 4.3 regarding the relationship between engagement and payment probability.

4.4.3 Monetization Dashboard

To visualize the predictive relationship between engagement level and subscription probability, a logistic curve was constructed based on the regression results presented in Section 4.3.

The X-axis represents the number of insight_view interactions, and the Y-axis shows the estimated probability of subscription.

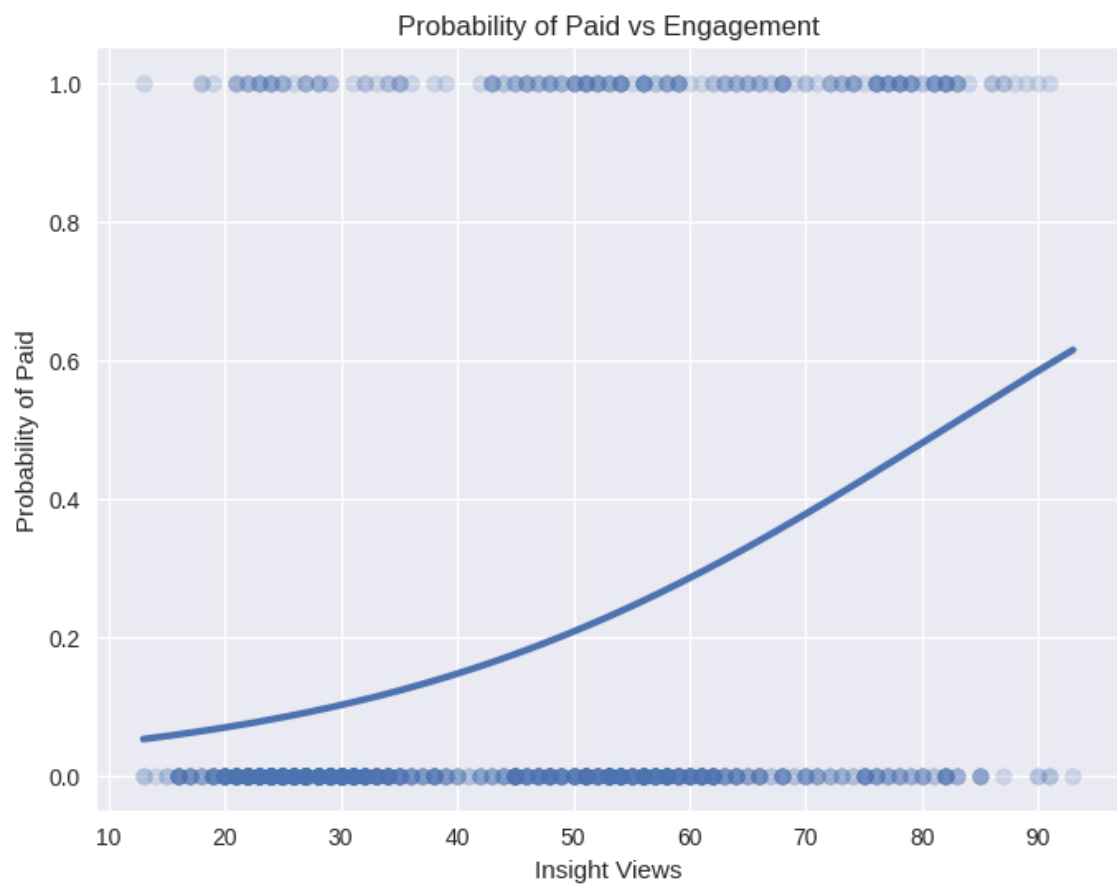


Figure 10. Subscription Probability as a Function of Engagement Level

Interpretation

As shown in Figure 10, the relationship between engagement and subscription probability is nonlinear and follows an S-shaped logistic curve.

At low interaction levels (up to approximately 30–40 views), subscription probability remains minimal.

At moderate engagement levels, probability increases more rapidly, indicating the presence of a behavioral threshold.

At high engagement levels, conversion probability rises significantly, confirming the hypothesis that depth of product usage plays a critical role in monetization.

This visualization demonstrates the practical value of predictive analytics in product management and can be used for:

- user segmentation based on payment probability
- trigger-based subscription scenarios
- optimization of paywall timing
- personalized offer generation

5 Evaluation of the Developed Framework

This section evaluates the proposed product analytics framework from three perspectives: technical architecture, analytical effectiveness, and operational applicability.

The objective is to assess whether the designed system ensures reliable metric calculation, supports product decision-making, and remains scalable for further development.

5.1 Technical Evaluation

The technical evaluation focuses on the robustness, scalability, and architectural consistency of the developed analytical system.

Architectural Consistency

The framework follows a layered architecture:

RAW → STAGING → INTERMEDIATE → MART → DASHBOARD

This separation ensures:

- isolation of raw data from business logic
- reproducibility of metric calculations
- clear transformation dependencies
- maintainability of analytical models

The use of Snowflake as a centralized data warehouse provides scalable storage and efficient query execution for large event datasets.

dbt introduces structured transformation logic through DAG-based modeling. Each transformation layer has a defined responsibility, which reduces coupling between components and simplifies debugging.

Scalability

The event-based model is inherently scalable.

New events, features, or metrics can be added without restructuring the entire system. Additional marts can be introduced without modifying upstream logic.

The modular dbt structure supports:

- extension of transformation layers
- incremental model execution
- parallel development

Thus, the framework can scale both in data volume and analytical complexity.

Reproducibility and Transparency

All metric calculations are implemented as SQL models within dbt.

This ensures:

- version control of transformations
- reproducibility of results
- traceability of dependencies

The DAG structure provides full visibility of metric lineage from raw events to final dashboards.

Data Quality Control

Built-in dbt testing mechanisms ensure:

- uniqueness of primary identifiers
- absence of null values in mandatory fields
- referential integrity between tables

The introduction of event naming conventions and schema governance further reduces the risk of analytical inconsistencies.

Overall, from a technical perspective, the framework demonstrates architectural robustness, scalability, and controlled transformation logic.

5.2 Analytical Effectiveness

The analytical evaluation focuses on whether the framework enables meaningful interpretation of user behavior and supports strategic product decisions.

Metric Coherence

The framework integrates activation, retention, engagement, monetization, and the North Star metric into a unified measurement system.

The selection of Weekly Insight Active Users (WIAU) as the North Star metric aligns with the core product value — meaningful interaction with personalized health analytics.

Supporting metrics such as retention, conversion, and subscription renewal provide contextual interpretation of performance dynamics.

Descriptive and Predictive Capabilities

The framework supports:

- descriptive analytics (WAU, retention, conversion)
- explanatory analysis (engagement differences between Free and Paid users)
- predictive modeling (logistic regression for payment probability)

This multi-level analytical capability demonstrates that the system goes beyond KPI reporting and enables behavioral interpretation.

Alignment with Product Logic

The event model reflects the actual user journey within a wearable-integrated application.

Engagement is defined not by passive data collection but by active interaction with analytics. This ensures that metrics represent real product value rather than technical activity.

Therefore, from an analytical standpoint, the framework provides structured, interpretable, and strategically relevant measurement.

5.3 Operational Effectiveness

Operational effectiveness evaluates the framework's applicability in a real product environment.

Decision Support

The dashboard layer allows monitoring:

- user growth and stability
- depth of core feature usage
- monetization dynamics
- behavioral thresholds

Managers can identify performance changes and assess the impact of product updates.

Early Signal Detection

Engagement metrics function as leading indicators of monetization.

This enables:

- timely subscription offer optimization
- proactive churn prevention
- targeted user segmentation

The predictive relationship between insight interactions and payment probability can inform trigger-based product strategies.

Governance and Maintainability

The defined naming conventions, schema governance, and layered modeling approach ensure:

- consistency across analytical models
- ease of onboarding new team members

- structured extension of the system

Thus, the framework is operationally viable and suitable for real-world implementation.

6 Conclusion

This work presented the design and evaluation of a product analytics framework for a wearable-integrated digital health application.

The developed system is based on an event-driven data model, layered analytical architecture, and structured transformation logic implemented through Snowflake and dbt.

The practical implementation using synthetic data demonstrated:

- stable user engagement dynamics
- strong correlation between engagement and monetization
- feasibility of predictive modeling
- support for data-driven decision-making

The proposed North Star metric — Weekly Insight Active Users (WIAU) — reflects meaningful user interaction with personalized analytics and aligns with long-term product value.

The evaluation confirmed that the framework:

- ensures technical robustness
- supports descriptive and predictive analytics
- enables operational decision support

Overall, the developed framework provides a scalable and reproducible analytical foundation for managing digital health products and can be adapted to other subscription-based digital platforms.

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